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ARC-6 FUSION ARCHITECTURE SPECIFICATION

Integrated Blueprint Reference Edition

Observer Strategic Computation Initiative / Arc-6 Complex

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FIELD	ENTRY
DOCUMENT ID	OSCI-ARC6-ARCH-FUS-2041-0001-TS / BPSPEC-INTEGRATED
ISSUING OFFICE	Office of Strategic Computation Infrastructure
PROGRAM	Observer Strategic Computation Initiative
DIVISION	Architecture Division
FACILITY	Arc-6 Complex
DOCUMENT TYPE	Fusion Architecture Specification with Blueprint Reference Package
SOURCE DRAWING PACKAGE	OSCI-ARC6-BPSPEC-2041-0003-TS; DWG-2041-0001-TS through DWG-2041-0002-TS
DATE OF ISSUE	14 March 2041
HANDLING	EYES ONLY - AUTHORIZED RECIPIENTS ONLY

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1.0 - PURPOSE

This document provides the complete architectural specification for the Arc-6 Fusion Complex, now expanded to include the blueprint reference package for formal drawing production.

- Core array layout
- Confinement coil architecture
- Arc Field coherence envelope
- Cryogenic routing
- Structural load distribution
- AGI substrate interface points
- Site plan geometry and Datacenter Monolith blueprint data

All architectural descriptions are grounded in real fusion research, high-field superconducting coil design, cryogenic engineering, plasma chamber geometry, and formal drawing conventions.

2.0 - FACILITY OVERVIEW

Arc-6 is a vertically stacked fusion and computation architecture consisting of eight high-field fusion cores, a central Arc Field generation lattice, cryogenic routing shafts, structural load rings, and Observer AGI substrate floors.

3.0 - FUSION CORE ARRAY ARCHITECTURE

3.1 - Core Arrangement

Arc-6 employs an octagonal core array arranged around the central Arc Field lattice.

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+---------------------------+
|       ARC-6 CORE ARRAY    |
+---------------------------+
| C1    C2    C3    C4      |
|       [ARC FIELD]         |
| C8    C7    C6    C5      |
+---------------------------+
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3.2 - Plasma Chamber Geometry

Each core uses a high-field tokamak geometry, based on real designs such as ITER and SPARC.

3.3 - Confinement Coils

- REBCO high-temperature superconducting coils
- 20-23 Tesla peak field strength
- Stable confinement at 120-150 million C

4.0 - ARC FIELD COHERENCE ENVELOPE

The Arc Field is a proprietary OSCI magnetic-photonic lattice that stabilizes AGI substrate coherence.

- 64 photonic lattice emitters
- 8 superconducting alignment rings
- 1 central coherence spine

5.0 - CRYOGENIC ROUTING ARCHITECTURE

- HTS coil temperature of 4-6 K
- DEC grid cooling
- Helium-3 storage stability

```
                    CRYOGENIC PLANT (SL-4)
                              v
PRIMARY HELIUM LINES -> HTS COIL JACKETS -> DEC GRID COOLING
              ^                                   |
              +------------- RETURN LINES ------------+
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6.0 - STRUCTURAL & MATERIALS ENGINEERING

Arc-6 uses tungsten shielding for D-D mode, graphene-reinforced composite mounts, niobium-tin superconducting bus supports, and load rings for coil, shielding, cryogenic jacket, and DEC grid assemblies.

7.0 - AGI SUBSTRATE INTERFACE POINTS

Arc-6 integrates directly with the Observer AGI substrate via photonic coherence channels, magnetic isolation corridors, substrate stabilization nodes, the coherence spine, photonic lattice emitters, and DEC grid telemetry.

8.0 - BLUEPRINT REFERENCE PACKAGE

This section integrates the missing dimensional, spatial, structural, notation, and drawing-production data from the Arc-6 Complex Blueprint Specification Package.

8.1 - Drawing Package Scope

- Drawing 1: OSCI-ARC6-DWG-2041-0001-TS - Arc-6 Complex Site Plan, top-down ground level and below-grade overview.
- Drawing 2: OSCI-ARC6-DWG-2041-0002-TS - Datacenter Monolith Cross-Section, east-west vertical section including full height and sublevels.
- All dimensions are in meters unless otherwise noted. Elevations reference Grade Level Datum (GLD) = +/-0.000 m.

8.2 - Drawing Standards and Sheet Format

ATTRIBUTE	SPECIFICATION
Standards	ANSI/ASME Y14.5-2018; ANSI/ASME Y14.100-2017; DoD MIL-STD-100G; ANSI Y14.2 line weights.
Sheet size	ANSI E, 34 x 44 in, or ISO A0, 841 x 1189 mm.
Title block	Bottom-right corner; 150 mm x 200 mm.
Projection	Third-angle orthographic projection.
Drawing 1 scale	1:5,000 recommended; minimum legible scale 1:10,000.
Drawing 2 scale	1:500 vertical and horizontal recommended; minimum 1:1,000.

LINE TYPE	USE	WEIGHT
Continuous Thick	Visible object edges	0.70 mm
Continuous Thin	Dimension and projection lines	0.25 mm
Dashed Thin	Hidden / below-grade features	0.35 mm
Chain Thin	Centerlines and axes	0.25 mm
Chain Thick	Cutting plane lines	0.70 mm
Dotted Thin	Boundary zones / field perimeters	0.25 mm
Phantom	Reference geometry / field lobes	0.25 mm

8.3 - Coordinate Reference and Site Geometry

PARAMETER	VALUE
Origin Point	POO = X 0.000, Y 0.000, Z 0.000; geometric center of Datacenter Monolith footprint at grade.
Axes	X east-west, east positive; Y north-south, north positive; Z vertical, up positive and down negative.
True North	007 degrees magnetic declination corrected; all Y-axis measurements are True North.
Arc Field Boundary	Radius 2,100.0 m from POO.
Security Perimeter	Radius 3,800.0 m from POO.

8.4 - Site Plan Element Specifications

ELEMENT	PRIMARY SPECIFICATION	DRAWING NOTES
A - Datacenter Monolith	Square footprint 180.0 m x 180.0 m; centered at POO; corners at +/-90.0 m; 1.800 m composite wall plus 0.050 m titanium cladding; 176.4 m interior clear dimension.	Continuous Thick; no fenestration; hexagonal exterior embossing.
B - Thermal Towers TT-1 to TT-4	Four circular towers; exterior diameter 28.0 m; wall thickness 1.200 m; height 180.0 m; centers at +/-102.700 m coordinates.	Continuous Thick outlines; Chain Thin centerlines; symbolic vapor discharge arcs.
C - North Access Corridor	Centerline X=0.000; exterior width 8.0 m; length 40.0 m; clear width 6.4 m; clear height 4.0 m; extends north from Y +90.0 to Y +130.0.	Four checkpoints: CP-1 Y +128.0, CP-2 +118.0, CP-3 +108.0, CP-4 +92.0.
D - Access Road	Centerline X=0.000; Y +130.0 to +800.0; 7.0 m paved width inside Ring-1; 5.0	Continuous Thin inside Ring-

	m controlled single lane Ring-1 to Ring-3.	1; Dashed Thin with Chain centerline beyond.
E - Ground-Loop Heat Exchanger	Below-grade loop around Monolith; symbolic circle r=140.0 m; depth 6.0 m to 12.0 m.	Dashed Thin; orange coding; annotate below-grade depth.
F - Compute Ring 1	Radius 200.0 m; 40 bunkers; 9.0 degree spacing; each bunker 12.0 m x 8.0 m; 4.0 m below grade.	CB-01 at X 0.000, Y +200.000; radial photonic fiber conduit.
G - Compute Ring 2	Radius 450.0 m; 24 hardened facilities; 15.0 degree spacing; each facility 24.0 m x 18.0 m.	Semi-submerged; 2.0 m above grade and 3.0 m below.
H - Compute Ring 3	Radius 800.0 m; 12 sensor tower clusters; 30.0 degree spacing; each tower base 6.0 m x 6.0 m; height 22.0 m.	Foundation depth 4.5 m; full ST-01 to ST-12 coordinates defined by formula.
I - Sublevel Fusion Deck	Hexagonal six-core array at Z -80.000 m; array diameter 400.0 m; core radius from center 200.0 m; individual SIV circle r=22.0 m; Crown r=30.0 m.	Dashed Thin amber; depth callout EL -80.000 M.
J - Arc Field Boundary	Circle r=2,100.0 m centered at POO; six lobe axes aligned to bearings 000, 060, 120, 180, 240, 300 degrees.	Phantom violet line; note civilian exclusion zone r=14,200 m not shown.
K - Security Perimeter	Radius 3,800.0 m from POO.	Represented by corner indicators at drawing scale.
L - North Arrow and Scale Bar	True North arrow; 0 to 1,000 m scale bar divided into 10 x 100 m segments.	Annotate: drawing scale 1:5,000; do not scale if reproduced.

8.5 - Coordinate Formulas and Selected Coordinates

SYSTEM	FORMULA / COORDINATES
Ring-1 bunkers	X(n)=200.0*sin(n*9.0 deg), Y(n)=200.0*cos(n*9.0 deg), n=0..39. Reference: CB-01 0/+200; CB-11 +200/0; CB-21 0/-200; CB-31 -200/0.
Ring-2 facilities	X(n)=450.0*sin(n*15.0 deg), Y(n)=450.0*cos(n*15.0 deg), n=0..23. Reference: HF-01 0/+450; HF-07 +450/0; HF-13 0/-450; HF-19 -450/0.
Ring-3 sensor towers	X(n)=800.0*sin(n*30.0 deg), Y(n)=800.0*cos(n*30.0 deg), n=0..11. ST-01 0/+800; ST-04 +800/0; ST-07 0/-800; ST-10 -800/0.
Fusion core centers	Alpha 0/+200/-80; Beta +173.205/+100/-80; Gamma +173.205/-100/-80; Delta 0/-200/-80; Epsilon -173.205/-100/-80; Zeta -173.205/+100/-80; Crown 0/0/-80.
YBCO trunk penetrations	Trunk-1 0/+4.500; Trunk-2 +3.897/+2.250; Trunk-3 +3.897/-2.250; Trunk-4 0/-4.500; Trunk-5 -3.897/-2.250; Trunk-6 -3.897/+2.250.

9.0 - DATACENTER MONOLITH CROSS-SECTION

9.1 - Overall Dimensions

ATTRIBUTE	VALUE
Cross-section	East-west vertical cut at Y=0, looking north.
Height above grade	220.0 m; roof at Z +220.000 m.
Below-grade sublevel depth	35.5 m; bottom of SL-4 floor at Z -35.500 m.
Fusion deck centerline	Z -80.000 m in bedrock.
Total roof-to-fusion-deck span	300.0 m.
Building width	180.0 m, X -90.0 to X +90.0.
Foundation	Mat foundation at Z -6.000 m; 1.500 m post-tensioned concrete; 180.0 m x 180.0 m.
Floor system	0.350 m post-tensioned flat plate; 9.0 m x 9.0 m column grid; typical 5.500 m floor-to-floor; 5.150 m clear ceiling.

9.2 - Vertical Zone Summary

LEVEL RANGE	ZONE	PRIMARY FUNCTION
SL-4 to SL-1	Sublevels	Cryogenic production, conduit origin, QAC control rooms, emergency systems, blast-rated power routing and manual overrides.
G to FL-03	Operational Base / HOC	Security corridor, logistics, life-support plant, 12-person Human Operations Center, communications hub, emergency coordination.
FL-04 to FL-10	Primary Compute Layer A	Neuromorphic silicon-photonic arrays, Tier-1 inference hardware, photonic switching, Layer A-B transition.
FL-11 to FL-18	Primary Compute Layer B	Recursive self-improvement substrate rows, RIC execution nodes, upper Faraday isolation boundary.
FL-19 to FL-24	Memory Architecture	Holographic memory banks, temporal storage tiers, archival systems, temporal coherence management.
FL-25 to FL-30	Control and Safety Systems	AGI safety co-processors, FSC/PMS hardware, Arc Field monitoring, oversight terminals, EIV network control.
FL-31 to FL-36	Power Distribution and Thermal Management	YBCO terminal nodes, switchgear, heat exchanger plant, thermal tower conduits, energy metering.
FL-37 to FL-40	The Apex / Roof	Remote manipulator arrays, orbital uplink equipment, restricted FL-39, roof/mechanical penthouse.

9.3 - YBCO Conduit Trunks

ATTRIBUTE	VALUE
Quantity	6 trunks.
Exterior diameter	2.400 m each.
Triple-layer jacket	0.300 m cryogenic insulation.
YBCO conductor inner diameter	1.800 m.
Vertical extent	Z -27.000 m, SL-3 origin, to Z +170.500 m, Floor 31 terminal nodes.
Total vertical run	197.500 m per trunk.
EIV positions	Z -27.000, +23.000, +73.000, +123.000, +170.500 m.

9.4 - Thermal Towers and Sublevel Fusion Deck

SYSTEM	SPECIFICATION

Thermal towers	TT-1 west and TT-2 east visible in E-W section; exterior diameter 28.0 m; wall thickness 1.200 m; height Z 0.000 to +180.000; foundation depth Z -8.000 to 0.000; centers at X +/-102.700 m.
Tower bridge	Structural and thermal duct bridge at Floor 33, Z +181.500 to +182.000; height 3.000 m; width 5.000 m; length 12.700 m.
Sublevel fusion deck	Core centerline Z -80.000 m; six 3.000 m rock-bored conduit shafts from Crown level to SL-3; shaft length 53.000 m.
Crown section representation	X=0, Z -80.000; circle r=30.0 m; dashed amber; annotated as YBCO superconducting ring, D=60.0 m.

9.5 - Structural Systems

SYSTEM	VALUE
Superstructure	Reinforced post-tensioned concrete core-and-frame; 40.0 m x 40.0 m central shear core; perimeter steel moment frame at 9.0 m grid.
Substructure	600 mm bored concrete piles at 3.0 m grid; bedrock approx. Z -25.000 to -30.000 m subject to geotechnical survey.
Seismic isolation	Triple Friction Pendulum isolators at each column base; isolation layer Z -2.000 m; moat clearance 0.800 m; design period 5.0 s; maximum credible displacement +/-0.650 m.
Faraday Deck 1	Z +60.000 to +60.500 m, between FL-10 and FL-11; post-tensioned concrete with copper mesh at 100 mm spacing; >80 dB attenuation above 100 MHz.
Faraday Deck 2	Z +99.000 to +99.500 m, between FL-18 and FL-19; same construction and isolation performance.

10.0 - RECOMMENDED ADDITIONAL DRAWING SECTIONS

DRAWING NO.	DESCRIPTION	VIEW	PRIORITY
DWG-2041-0003-TS	Site Plan - Sublevel Fusion Deck Only	Plan at Z -80.000 m	High
DWG-2041-0004-TS	Monolith Cross-Section, N-S at X=0	Vertical Section	High
DWG-2041-0005-TS	Sublevel Sections SL-1 through SL-4	Plan at each sublevel	High
DWG-2041-0006-TS	Crown and Conduit Nexus Detail	Section and Plan	High
DWG-2041-0007-TS	Typical Fusion Core SIV Detail	Section and Plan	High
DWG-2041-0008-TS	Thermal Tower Detail Section	Vertical Section	Medium
DWG-2041-0009-TS	Faraday Isolation Deck Detail	Detail Section	Medium
DWG-2041-0010-TS	HOC Floor Plan, Floor 03	Plan	Medium
DWG-2041-0011-TS	Apex Floor Plans, Floors 37-40	Plan each level	High
DWG-2041-0016-TS	YBCO Conduit Trunk Typical Section	Detail Section	High
DWG-2041-0017-TS	EIV Assembly Detail	Detail Section	Medium

11.0 - SUMMARY DIMENSIONS

ELEMENT	DIMENSION	VALUE
Datacenter Monolith	Plan width/depth	180.000 m x 180.000 m
Datacenter Monolith	Height above grade	220.000 m
Datacenter Monolith	Sublevel depth	35.500 m
Datacenter Monolith	Total floors	40 above grade + 4 sublevels = 44 total
Fusion Deck	Centerline depth	Z -80.000 m
Arc Field Boundary	Radius	2,100.0 m
Security Perimeter	Radius	3,800.0 m
Ring-1	Radius / units	200.0 m / 40 bunkers
Ring-2	Radius / units	450.0 m / 24 hardened facilities

Ring-3	Radius / units	800.0 m / 12 sensor tower clusters

12.0 - CITATIONS AND SOURCES

Real Scientific and Engineering Sources

- ITER Organization, Tokamak Engineering Design, 2023.
- MIT Plasma Science & Fusion Center, REBCO Coil Demonstration, Nature 2021.
- LLNL Cryogenic Systems Group, Helium Loop Stability, 2022.
- Fusion Engineering and Design Journal, Structural Load Management, 2020.
- ANSI/ASME Y14.5-2018; ANSI/ASME Y14.100-2017; ANSI Y14.2 line conventions.

Classified OSCI Sources (Fictional)

- OSCI Internal Memo 14-A, Arc Field Coherence Envelope, 2040.
- OSCI Substrate Diagnostics Report 3-B, Observer-3 Coherence Drift Analysis, 2041.
- OSCI Cryogenic Routing Architecture Brief, 2039.
- OSCI-ARC6-BPSPEC-2041-0003-TS, Arc-6 Complex Blueprint Specification Package, Rev. 0.

13.0 - DOCUMENT CONTROL RECORD

FIELD	ENTRY	STATUS
COPY 01	Director, Office of Strategic Computation Infrastructure	ACTIVE
COPY 02	Arc-6 Complex Command Archive	SEALED
COPY 03	Observer Strategic Computation Initiative / Architecture Division	ACTIVE
COPY 04	Master Control AI Oversight Cell	ACTIVE

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